

Exercise sheet 3 : The Black-Scholes Model

The notations are the same as in the course. In all exercises, we consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F})$ as well as an $\mathbb{F}$ -Brownian motion $(B_t)_{t \geq 0}$.

Exercise 1. In the Black-Scholes model, a self-financing strategy is described by a pair $(x, \theta) \in \mathbb{R} \times \mathcal{A}$. Let $V_t^{x, \theta}$ denote the value of this strategy at time t , and $\tilde{V}_t^{x, \theta}$ its discounted value.

1. Give $\tilde{V}_t^{x, \theta}$ as a function of x and θ . Compute $\mathbb{E}[\tilde{V}_t^{x, \theta}]$ and $\mathbb{V}(\tilde{V}_t^{x, \theta})$.
2. Justify that the probability measure $\mathbb{Q}$ defined by

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left(-\frac{1}{2} \left(\frac{\mu - r}{\sigma} \right) t - \frac{\mu - r}{\sigma} B_t \right)$$

is a risk-neutral probability measure.

Exercise 2. Consider a call option on the underlying S , with maturity T and strike K . Its terminal value is given by the random variable $G = (S_T - K)_+$.

1. Let $u(t, x) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[G | S_t = x]$. Express $u(t, x)$ as a function of t, x, r, σ , and K and the cumulative distribution function N of a $\mathcal{N}(0, 1)$ distribution.
2. Compute the probability under $\mathbb{P}$ and under $\mathbb{Q}$ that the option is exercised.
3. Verify that u is a $\mathcal{C}^{1,2}$ function on $[0, T[\times \mathbb{R}_+^*$. Determine the expressions of

$$\Delta(t, x) = \partial_x u(t, x), \quad \Gamma(t, x) = \partial_{xx}^2 u(t, x), \quad \theta(t, x) = \partial_t u(t, x).$$

Show that the function u satisfies

$$\begin{cases} ru(t, x) = \partial_t u(t, x) + rx\partial_x u(t, x) + \frac{1}{2}\sigma^2 x^2 \partial_{xx}^2 u(t, x) & \forall (t, x) \in [0, T[\times \mathbb{R}_+^* \\ u(T, x) = (x - K)_+ & \forall x \in \mathbb{R}_+^*. \end{cases}$$

4. Consider the process $\theta^* = (\partial_x u(t, S_t))_{t \in [0, T]}$. Show that $\theta^* \in \mathcal{A}$.
5. Consider the self-financing strategy $(u(0, S_0), \theta^*)$. Show that this strategy replicates the call option. Justify then that the price at time t of the call option is given by $p_t = u(t, S_t)$.

Exercise 3. Consider a contingent claim with payoff $G \geq 0$ almost surely such that $\mathbb{E}^{\mathbb{Q}}[G^2] < +\infty$ and with maturity $T > 0$. Show that it is not possible to replicate G using a strategy satisfying $\theta_t = \frac{c}{S_t}$ for some $c \in \mathbb{R}^*$.

Exercise 4. Consider a call option on the underlying S , with maturity T and strike K . Its terminal value is given by the random variable $G = (S_T - K)_+$. Suppose that the price of the underlying follows the dynamics

$$\frac{dS_t}{S_t} = \mu dt + \sigma_t dB_t,$$

where $(\sigma_t)_{t \in [0, T]} \in L^2([0, T])$ is deterministic, positive, and never vanishes.

1. Find the formula for the option price, and show that the price at time $t = 0$ is equal to the price of a European call option on an underlying with constant volatility to be specified.
2. Give the formula for Δ and θ .
3. Show that if $(\sigma_t)_{t \in [0, T]}$ is not constant, the option price over the interval $[0, T]$ does not correspond to the price of a European call option on an underlying with constant volatility.

Exercise 5. Consider, with the notations from the lecture, an Asian call option with payoff G , maturity T , and strike K on an underlying S whose price follows a Black-Scholes dynamics, i.e.,

$$G = \left(\frac{1}{T} \int_0^T S_t dt - K \right)_+.$$

Show that its price is lower than that of a European call option with the same characteristics.

Exercise 6. Let $T > 0$ and $C^E(t, S, K)$ be the price at time $t \in [0, T]$ of a European call option with maturity T and strike K when the underlying price is S . For any $K > 0$, we introduce another derivative product, called a digital call, with payoff $G = \mathbb{1}_{\{S_T \geq K\}}$; its price will be denoted $C^D(t, S, K)$. Suppose that for all $t \in [0, T]$, C^D is continuous with respect to K , and C^E is $\mathcal{C}^1$ with respect to K .

1. For any K and $\Delta K > 0$, consider a portfolio with value Π_t^1 containing ΔK digital calls with strike K and one European call option with strike $K + \Delta K$. Also, consider a portfolio with value Π_t^2 containing ΔK digital calls with strike $K + \Delta K$ and one European call option with strike $K + \Delta K$. Show that for all $T \in [0, T]$, we have

$$\Pi_t^2 \leq C^E(t, S, K) \leq \Pi_t^1.$$

2. Deduce that $C^D(t, S, K) = -\frac{\partial C^E(t, S, K)}{\partial K}$.
3. If the underlying asset price S follows the Black-Scholes model, find explicitly the strategy x^G, θ^G that replicates the payoff G .

Exercise 7. Let $T > 0$ and $C^E(t, S, K)$ be the price at time $t \in [0, T]$ of a European call option with maturity T and strike K when the underlying price is S . For any $K > 0$, we introduce another derivative product, called a digital call, with payoff $G = \mathbb{1}_{\{S_T \geq K\}}$; its price will be denoted $C^D(t, S, K)$. Suppose that for all $t \in [0, T]$, C^D is continuous with respect to K , and C^E is $\mathcal{C}^1$ with respect to K .

1. Calculate $C^D(t, S, K)$ and $\frac{\partial C^D(t, S, K)}{\partial S}$.
2. Give the price of an option with payoff $S_T \mathbb{1}_{\{S_T \geq K\}}$.
3. Can the digital call be replicated with a portfolio that does not contain any other risky assets than the European call?

Exercise 8. We consider the Black-Scholes model. Let $T > 0$ and $T_1 \in]0, T[$. For any $T \in [0, T]$, let $C_t(S, T, K)$ be the price at time t of a European call option on the underlying S , with maturity T and strike $K > 0$. Let G be a European call option with strike $K_1 > 0$ and maturity T_1 , where the underlying is a European call option on S with maturity T , i.e.,

$$G = (C_{T_1}(S, T, K) - K_1)_+.$$

1. Show that for any $t \in [0, T_1]$, the value of G at time t is given by $v(T_1 - t, S_t)$, where

$$v(\tau, x) = e^{-r\tau} \mathbb{E} \left[\left(u \left(T_1, x e^{\left(r - \frac{\sigma^2}{2}\right)\tau + \sigma\sqrt{\tau}Z} \right) - K_1 \right)_+ \right],$$

where Z is a random variable with distribution $\mathcal{N}(0, 1)$ and $u(t, x)$ is the price at time t of the European call option on S , with maturity T and strike K , when $S_t = x$.

2. Show that

$$v(\tau, x) = e^{-r\tau} \mathbb{E} \left[u \left(T_1, x e^{\left(r - \frac{\sigma^2}{2}\right)\tau + \sigma\sqrt{\tau}Z} \right) \mathbf{1}_{\{Z > -d\}} \right] - e^{-r\tau} K_1 N(d),$$

with

$$d = \frac{\ln\left(\frac{x}{x_1}\right) - \left(r - \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}}$$

where x_1 is the unique solution of the equation $u(T_1, x) = K_1$.

3. Show that by taking $\tau = T_1 - t$, we have

$$v(\tau, x) + e^{-r\tau} K_1 N(d) = \mathbb{E} \left[\left(x e^{\sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1) - \frac{\sigma^2}{2}(\tau + \tau_1)} - K e^{-r(\tau + \tau_1)} \right) \mathbf{1}_A \right],$$

where $\tau_1 = T - T_1$, Z_1 is a random variable with distribution $\mathcal{N}(0, 1)$ independent of Z , and A is given by

$$A = \left\{ \sigma(\sqrt{\tau}Z + \sqrt{\tau_1}Z_1) > - \left(\ln\left(\frac{x}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(\tau + \tau_1) \right) \right\} \cap \{Z > -d\}.$$

4. Deduce an expression for $v(T_1 - t, x)$ using N and N_2 defined by

$$N_2(y, y_1, \rho) = \mathbb{P}(Z < y, Z + \rho Z_1 < y_1).$$

5. Show that this option can be replicated using a portfolio containing only the risk-free asset and the risky asset S .